

A 30-Dimensional Candidate Counterexample to Pierce–Birkhoff

A symmetric-matrix simplification of the two-matrix construction

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Abstract

We record a proposed reduction of a two-matrix Pierce–Birkhoff counterexample from $M_6(\mathbb{R}) \times M_6(\mathbb{R}) \cong \mathbb{R}^{72}$ to $\text{Sym}_5(\mathbb{R}) \times \text{Sym}_5(\mathbb{R}) \cong \mathbb{R}^{30}$. The construction is a continuous positively homogeneous piecewise quadratic function whose active branch is gated by the sign of $\det X$. The proof has four ingredients: a homogeneity reduction from arbitrary polynomial lattice expressions to finitely many degree-at-most-two sign tests; a diagonally-dominant quadratic barrier; a low-degree orbit lemma; and two perturbations that have identical quadratic product data but opposite determinant signs. A final section explains an important caveat: the most obvious infinitesimal $+/-$ pair does not itself witness failure of the separating-ideal criterion, because cubic minors already separate the two arcs at lower infinitesimal order.

1 The proof architecture

The construction isolates the following mechanism.

1. **Homogeneity truncates lattice information.** If a function g satisfies $g(tz) = t^2g(z)$ for every $t > 0$ and has a finite lattice expression in arbitrary real polynomials, then the sign of g is determined by the signs of finitely many polynomials of degree at most two.
2. **A quadratic barrier creates a safe region.** We build a piecewise quadratic h such that $h > 0$ forces XY , hence X , to be invertible. The sign of $\det X$ can then be used as a branch condition without creating a discontinuity inside $\{h > 0\}$.
3. **A degenerate orbit has very few quadratic equations.** On a suitable orbit $\mathcal{O}$, one has $XY = 0$. We prove that every polynomial of degree at most two vanishing on $\mathcal{O}$ is a linear functional of XY .
4. **The hidden sign is inserted in a common null direction.** Two perturbations approach the same point of $\mathcal{O}$, satisfy $XY = \varepsilon^2 I$ exactly, but have opposite signs of $\det X$. Quadratic tests cannot detect the hidden sign, while the determinant does.

The dimension $5 = 2 + 2 + 1$ is natural for this mechanism: two rank-two supports are kept disjoint, and one additional common null direction is used to store a sign.

2 The 30-dimensional function

Let

$$V = \text{Sym}_5(\mathbb{R}) \times \text{Sym}_5(\mathbb{R}), \quad \dim V = 15 + 15 = 30.$$

Write a point of V as (X, Y) and put

$$P = XY.$$

Although X and Y are symmetric, P need not be symmetric.

For $i \neq j$ and $\tau \in \{+1, -1\}$ define the homogeneous quadratic forms

$$q_{ij,\tau}(X, Y) = P_{ii} + 4\tau P_{ij}.$$

There are $5 \cdot 4 \cdot 2 = 40$ such forms. Set

$$h(X, Y) = \min_{i \neq j, \tau = \pm 1} q_{ij,\tau}(X, Y).$$

Finally define

$$f(X, Y) = \begin{cases} h(X, Y), & h(X, Y) > 0 \text{ and } \det X > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Proposition 1. *If $h(X, Y) > 0$, then $P = XY$ is invertible. In particular X and Y are invertible.*

Proof. For every $i \neq j$, positivity of both $q_{ij,+}$ and $q_{ij,-}$ gives

$$P_{ii} > 4|P_{ij}|.$$

Hence $P_{ii} > 0$ and

$$\sum_{j \neq i} |P_{ij}| < P_{ii}$$

for every row i . Thus P is strictly diagonally dominant by rows. If $Pv = 0$ and $|v_i| = \max_j |v_j| > 0$, then

$$P_{ii}|v_i| \leq \sum_{j \neq i} |P_{ij}| |v_j| \leq \left(\sum_{j \neq i} |P_{ij}| \right) |v_i| < P_{ii}|v_i|,$$

a contradiction. Therefore P is invertible. Since $P = XY$, both factors are invertible. $\square$

Proposition 2. *The function f is continuous, positively homogeneous of degree two, and piecewise polynomial with quadratic polynomial labels on a finite closed semialgebraic cover.*

Proof. On the open set $\{h > 0\}$, Proposition 1 gives $\det X \neq 0$, so $\text{sgn}(\det X)$ is locally constant. On $\{h < 0\}$, the function is locally zero. At $h = 0$ we have

$$0 \leq f(X, Y) \leq \max\{h(X, Y), 0\},$$

which proves continuity.

For the finite polynomial cover, enumerate the forty quadratics as $q_1, \dots, q_{40}$ and put

$$F_0 = \{\det X \leq 0\} \cup \bigcup_j \{q_j \leq 0\},$$

and for $1 \leq i \leq 40$,

$$F_i = \{\det X \geq 0, q_j \geq 0 \forall j, q_i \leq q_j \forall j\}.$$

These are closed semialgebraic sets covering V . Give F_0 the label 0 and F_i the label q_i . On F_i , $q_i = h \geq 0$. If $h > 0$, then X is invertible, so $\det X \geq 0$ implies $\det X > 0$ and $f = q_i$. On $F_i \cap F_0$, either some $q_j = 0$, hence $q_i = 0$, or $\det X = 0$; the latter cannot occur while every $q_j > 0$ by Proposition 1. Thus the labels agree on overlaps.

Finally, $q_{ij,\tau}(tX, tY) = t^2 q_{ij,\tau}(X, Y)$ and $\det(tX) = t^5 \det X$ for $t > 0$, hence

$$f(tX, tY) = t^2 f(X, Y).$$

□

3 Homogeneity reduces arbitrary lattice expressions to quadratic sign tests

A finite polynomial lattice expression is obtained from finitely many real polynomials using finitely many min and max operations.

Lemma 3 (Homogeneity reduction). *Let $g : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfy*

$$g(tz) = t^2 g(z) \quad (t > 0).$$

If g has a finite lattice expression in arbitrary real polynomials, then there exists a finite family of real polynomials of degree at most two such that any two points having the same signs for every polynomial in the family have the same sign of g .

Proof. Write

$$g = L(p_1, \dots, p_m),$$

where L is a fixed expression using only min and max. Decompose

$$p_j = c_j + \ell_j + Q_j + \sum_{d \geq 3} P_{j,d}$$

into homogeneous parts. Positive scaling commutes with min and max, hence

$$g(z) = L\left(\frac{p_1(tz)}{t^2}, \dots, \frac{p_m(tz)}{t^2}\right).$$

As $t \downarrow 0$, each entry has an extended-real limit

$$u_j(z) = \begin{cases} \operatorname{sgn}(c_j) \infty, & c_j \neq 0, \\ \operatorname{sgn}(\ell_j(z)) \infty, & c_j = 0, \ell_j(z) \neq 0, \\ Q_j(z), & c_j = 0, \ell_j(z) = 0. \end{cases}$$

Terms of degree at least three disappear after division by t^2 . Since finite min and max are continuous on the extended real line,

$$g(z) = L(u_1(z), \dots, u_m(z)).$$

The sign map on $[-\infty, \infty]$ is nondecreasing and therefore preserves min and max. Thus $\operatorname{sgn} g(z)$ is determined by the signs of the finitely many ℓ_j and Q_j . □

Therefore, to rule out every polynomial lattice representation of f , it is enough to show that no finite family of degree-at-most-two sign tests determines $\operatorname{sgn} f$.

4 The boundary orbit and its quadratic equations

Define

$$D = \text{diag}(1, 1, 0, 0, 0), \quad E = \text{diag}(0, 0, 1, 1, 0),$$

and consider the orbit

$$\mathcal{O} = \left\{ \left(UDU^T, U^{-T}EU^{-1} \right) : U \in GL_5(\mathbb{R}) \right\} \subset V. \quad (2)$$

For every $(X, Y) \in \mathcal{O}$,

$$XY = UDU^T U^{-T} EU^{-1} = UDEU^{-1} = 0,$$

and similarly $YX = 0$.

Lemma 4 (Quadratic orbit lemma). *A real polynomial $p(X, Y)$ of total degree at most two vanishes on $\mathcal{O}$ if and only if*

$$p(X, Y) = L(XY)$$

for some real linear functional L on $M_5(\mathbb{R})$.

Proof. The reverse implication is immediate from $XY = 0$ on $\mathcal{O}$.

For the converse, first note that replacing U by

$$U \text{ diag}(a, a, b^{-1}, b^{-1}, 1)$$

replaces (X, Y) by (a^2X, b^2Y) . Hence the bihomogeneous components of p in (X, Y) vanish separately on $\mathcal{O}$.

There are no nonzero pure X components of degree one or two. Indeed, the X -projection of $\mathcal{O}$ is the congruence orbit of D , consisting of positive semidefinite symmetric matrices of rank two. Its Euclidean closure contains every rank-one matrix uu^T . A linear form vanishing on all uu^T is zero. If a homogeneous quadratic form Q vanishes on the rank-two positive semidefinite locus, then it also vanishes on uu^T and on $uu^T + vv^T$. Polarization gives

$$\tilde{Q}(uu^T, vv^T) = 0$$

for all u, v . Since rank-one symmetric matrices span $\text{Sym}_5(\mathbb{R})$, $Q = 0$. The same argument applies to pure Y components. The constant component is also zero. Thus only a bilinear form $B(X, Y)$ may remain.

We next show that the Euclidean closure of $\mathcal{O}$ contains every pair

$$(uu^T, ss^T) \quad \text{with } u^T s = 0.$$

At the standard point take

$$U_\delta = \text{diag}(1, \delta, 1, \delta^{-1}, 1).$$

Then

$$U_\delta D U_\delta^T = E_{11} + \delta^2 E_{22}, \quad U_\delta^{-T} E U_\delta^{-1} = E_{33} + \delta^2 E_{44},$$

which tends to (E_{11}, E_{33}) . If $u^T s = 0$, choose $W \in GL_5(\mathbb{R})$ with $W e_1 = u$ and $W^{-T} e_3 = s$; applying the fixed change of basis W gives the desired pair in the closure.

Hence

$$B(uu^T, ss^T) = 0 \quad \text{whenever } u^T s = 0.$$

Set

$$F(u, s) = B(uu^T, ss^T).$$

This polynomial has bidegree $(2, 2)$ and vanishes on the hypersurface $u^T s = 0$. The polynomial $u^T s$ is irreducible (its bilinear rank is greater than one), and its real zero set is Zariski dense in that hypersurface. Therefore

$$F(u, s) = (u^T s)G(u, s)$$

with G of bidegree $(1, 1)$. Thus $G(u, s) = u^T A s$ for some matrix A , so

$$F(u, s) = (u^T s)(u^T A s).$$

Define $L(M) = \text{tr}(A^T M)$. Since

$$(uu^T)(ss^T) = (u^T s)us^T,$$

we have

$$B(uu^T, ss^T) = L(uu^T ss^T).$$

Rank-one symmetric matrices span $\text{Sym}_5(\mathbb{R})$, so bilinearity extends the identity to all symmetric X, Y :

$$B(X, Y) = L(XY).$$

□

5 The two perturbations

Fix $U \in GL_5(\mathbb{R})$. For $\eta \in \{+1, -1\}$ and $\varepsilon > 0$, define

$$D_X^\eta = \text{diag}(1, 1, \varepsilon^2, \varepsilon^2, \eta\varepsilon),$$

$$D_Y^\eta = \text{diag}(\varepsilon^2, \varepsilon^2, 1, 1, \eta\varepsilon),$$

and

$$X_\eta = U D_X^\eta U^T, \quad Y_\eta = U^{-T} D_Y^\eta U^{-1}. \quad (3)$$

The design has three simultaneous properties.

A common limit

For both signs η ,

$$(X_\eta, Y_\eta) \longrightarrow (U D U^T, U^{-T} E U^{-1}) \in \mathcal{O} \quad (\varepsilon \downarrow 0).$$

Identical quadratic product data

Direct multiplication gives

$$D_X^\eta D_Y^\eta = \text{diag}(\varepsilon^2, \varepsilon^2, \varepsilon^2, \varepsilon^2, \eta^2 \varepsilon^2) = \varepsilon^2 I_5.$$

Hence

$$X_\eta Y_\eta = \varepsilon^2 I_5, \quad Y_\eta X_\eta = \varepsilon^2 I_5. \quad (4)$$

The sign η has been squared away.

Opposite determinant signs

On the other hand,

$$\det D_X^\eta = \eta \varepsilon^5,$$

so

$$\det X_\eta = (\det U)^2 \eta \varepsilon^5. \quad (5)$$

Therefore

$$\det X_+ > 0, \quad \det X_- < 0.$$

The determinant remembers the hidden sign because it has odd degree in the fifth coordinate.

Since $X_\eta Y_\eta = \varepsilon^2 I$, all off-diagonal entries of P vanish and all diagonal entries equal ε^2 . Thus every defining quadratic satisfies

$$q_{ij,\tau}(X_\eta, Y_\eta) = \varepsilon^2,$$

and consequently

$$h(X_\eta, Y_\eta) = \varepsilon^2.$$

It follows from (1) and (5) that

$$f(X_+, Y_+) = \varepsilon^2, \quad f(X_-, Y_-) = 0. \quad (6)$$

6 Defeating an arbitrary finite quadratic sign family

Proposition 5. *For every finite family of real polynomials $p_1, \dots, p_M$ of degree at most two, there exist two points $z_+, z_- \in V$ having identical signs for all p_j , but satisfying*

$$f(z_+) > 0, \quad f(z_-) = 0.$$

Proof. Split the p_j into two classes.

If p_j does not vanish identically on $\mathcal{O}$, then its pullback

$$U \mapsto p_j(UDU^T, U^{-T}EU^{-1})$$

is a nonzero rational function on $GL_5(\mathbb{R})$. A finite collection of such nonzero rational functions is simultaneously nonzero at some $U \in GL_5(\mathbb{R})$. Fix such a generic U and write

$$o = (UDU^T, U^{-T}EU^{-1}).$$

Then $p_j(o) \neq 0$ for every polynomial in this first class. Since both perturbations (3) tend to o , continuity gives the same strict sign for $p_j(X_+, Y_+)$ and $p_j(X_-, Y_-)$ once ε is sufficiently small.

If p_j vanishes identically on $\mathcal{O}$, Lemma 4 gives

$$p_j(X, Y) = L_j(XY).$$

Using (4),

$$p_j(X_+, Y_+) = L_j(\varepsilon^2 I) = p_j(X_-, Y_-)$$

exactly.

There are finitely many p_j , so one sufficiently small $\varepsilon > 0$ works for all of them. Equation (6) then gives the required difference in f . $\square$

Combining Proposition 5 with Lemma 3 yields the advertised lattice obstruction.

Theorem 6 (The 30-dimensional lattice obstruction). *The function $f : \text{Sym}_5(\mathbb{R}) \times \text{Sym}_5(\mathbb{R}) \rightarrow \mathbb{R}$ defined in (1) is continuous, positively homogeneous of degree two, and piecewise polynomial on a finite closed semialgebraic cover with quadratic labels. Subject to the preceding lemmas, it admits no finite lattice expression in real polynomials of arbitrary degrees.*

7 Why dimension five is natural in this mechanism

The construction uses three blocks:

$$\underbrace{2}_{\text{support of } D} + \underbrace{2}_{\text{support of } E} + \underbrace{1}_{\text{common null direction}} = 5.$$

The rank-two supports are disjoint, which gives $DE = 0$. Rank two is also large enough that the pure rank locus has no nonzero equations of degree at most two. The fifth direction is zero for both limiting matrices and is therefore available to carry the sign η . In the perturbation, the same sign is inserted into both matrices:

$$(\eta\varepsilon)(\eta\varepsilon) = \varepsilon^2,$$

so quadratic product data forgets the sign while the odd-degree determinant of X retains it.

This explains both the reduction from 6×6 to 5×5 and the use of symmetric matrices. The latter reduces the ambient dimension from 50 to 30 and simplifies the quadratic orbit lemma substantially.

8 Separation-ideal caveat

The preceding global lattice argument should not be confused with a direct construction of a pair of real-spectrum points violating the local separating-ideal criterion. In fact, the most obvious infinitesimal version of the perturbation does *not* provide such a pair.

Replace ε by a positive infinitesimal t and first take $U = I$. Then

$$X_\eta = \text{diag}(1, 1, t^2, t^2, \eta t), \quad XY = t^2 I.$$

Let

$$r(X) = \det X_{\{1,2,5\},\{1,2,5\}}.$$

Then

$$r(X_+) = t, \quad r(X_-) = -t.$$

Thus the cubic polynomial r separates the two infinitesimal branches. A quadratic label q has value t^2 on both branches, so

$$(r + q)(X_+) = t + t^2 > 0, \quad (r + q)(X_-) = -t + t^2 < 0.$$

Hence both r and $r + q$ are separators. Since the separating ideal is an ideal,

$$q = (r + q) - r$$

belongs to it.

For generic U , the same phenomenon persists: suitable 3×3 minors have leading term ηct with $c > 0$. Therefore the natural $+/-$ perturbation pair is not itself a witness to failure of the local separation condition.

Remark 7 (Interpretation). *The quadratic-sign argument controls only degree-at-most-two tests. A separating ideal, however, is generated by sign-changing polynomials of arbitrary degree, and a higher-degree polynomial can have lower infinitesimal order along a chosen pair of arcs. Thus polynomial degree and valuation order are different filtrations. Any direct local counterexample must control the latter, not merely the former.*

This caveat does not identify a flaw in the global sign-family argument above; rather, it says that if the global argument is correct, the real-spectrum pair witnessing failure of the local criterion cannot simply be the naive infinitesimal realization of the displayed $+/-$ arcs.

Summary

The 30-dimensional construction can be compressed to the following diagram:

$$\begin{array}{l}
 V = \text{Sym}_5(\mathbb{R})^2, \quad h = \min_{i \neq j, \tau} (XY)_{ii} + 4\tau (XY)_{ij}, \\
 h > 0 \implies XY \text{ invertible}, \\
 f = h \cdot \mathbf{1}_{\{h > 0, \det X > 0\}}, \\
 \mathcal{O} = \{(UDU^T, U^{-T}EU^{-1})\}, \quad XY = 0 \text{ on } \mathcal{O}, \\
 I(\mathcal{O})_{\leq 2} = \{L(XY)\}, \\
 X_\eta = UD_X^\eta U^T, \quad Y_\eta = U^{-T}D_Y^\eta U^{-1}, \\
 X_\eta Y_\eta = \varepsilon^2 I, \quad \text{sgn det } X_\eta = \eta, \\
 f(X_+, Y_+) = \varepsilon^2, \quad f(X_-, Y_-) = 0.
 \end{array}$$

The construction therefore hides a determinant sign in a common null direction that every quadratic product test forgets.